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MILITARY SPECIFICATION

RECTIFIER, TRANSISTOR ASSEMBLY: 10525740

1. SCOPE

1.1 This specification covers one type of transistorized rectifier assembly known as Rectifier, Transistor Assembly: 10525740, hereinafter referred to as "rectifier assembly". The rectifier assembly is a circuit board which houses diode-rectifiers and transistors used in conjunction with other circuit boards to provide regulated power for the Gun Direction Computer: M18.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Munitions Command

F10525740	Rectifier, Transistor Assembly: 10525740

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PACKAGING DATA SHEET

M14002 Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The rectifier assembly shall be manufactured in accordance with Drawing F10525740 and drawings pertaining thereto.

3.2 General specifications.- The rectifier assembly shall meet the following requirements of MIL-F-13926 and MIL-C-46357.

3.2.1 MIL-F-13926

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Workmanship

3.2.2 MIL-C-46357

(a) Etched circuit assembly
(b) Marking
(c) Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The rectifier assembly shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the logic driver for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the rectifier assembly shall not be damaged or adversely affected and shall meet the requirements of 3.5.

3.3.2 Shock.- The rectifier assembly shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.5.

3.3.3 Temperatures.

3.3.3.1 Operating.- The rectifier assembly shall meet the requirements of 3.5.3 while exposed to temperatures of -40°F and +135°F.

3.3.3.2 Storage.- The rectifier assembly shall meet the requirements of 3.5.1 and 3.5.2 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80° and +160°F.

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3.3.4 Humidity.

3.3.4.1 Operating.- The rectifier assembly shall meet the requirements of 3.5.3 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of +135°F (only).

3.3.4.2 Non-operating.- The rectifier assembly shall meet the requirements of 3.5 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The rectifier assembly shall show no evidence of damage and shall meet the requirements of 3.5, after being inoculated with the composite spore suspension specified in MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The rectifier assembly shall show no evidence of physical damage and meet the requirements of 3.5 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Isolation.- Parts shall be electrically isolated from their heat sinks.

3.5 Performance.- Unless otherwise specified, the rectifier assembly shall meet the performance requirements of this specification at standard ambient temperatures between 60° and 90°F and at a barometric pressure of 28 to 32 inches of mercury with a RH up to 90%. A minimum of 300 feet perminute of air at the ambient temperature (imposed by requirement) shall flow across the components mounted on the circuit board.

3.5.1 Diodes.

3.5.1.1 Reverse current.- The leakage current (I_R) through diodes CR 1-2-10-11-18-20-21-23, shall be less than 6 milliamperes (ma) when each diodes is back-biased with 150 direct-current (Vdc) ± 2%.

3.5.1.2 Reverse current.- The leakage current (I_R) through diodes CR3, 4, 5, 6, 7, 8, 12, 13, 14, 15, 16, 17, 24, 25, 26, 27, 28 and 29, shall be less than 36 microamperes (ua), when each diode is back-biased with 250 Vdc ± 2%.

3.5.1.3 Forward voltage drop.- The forward voltage drop (V_F) across diodes CR 1, 2, 10, 11, 18, 20, 21 and 23, shall be less than 1.5 Vdc, when each diode has a forward current (I_F) of a 1 ampere (amp), nominal.

3.5.1.4 Forward voltage drop.- The forward voltage drop (V_F) across diodes CR 3, 4, 5, 6, 7, 8, 12, 13, 14, 15, 16, 17, 24, 25, 26, 27, 28 and 29 shall be less than 1.5 Vdc, when each diode has a forward current (I_F) of 200 ma (nominal).

3.5.1.5 Zener voltages (CR 19 and CR 22).- The zener voltage (Vz) of diodes CR 19 and CR 22 shall be 6.0 ± 0.36 Vdc, when each diode has a nominal current of 250 ma flowing through it.

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3.5.1.6 Zener voltage (CR 9).- The zener voltage (V_Z) of diode CR 9 shall be 33 ± 2 Vdc, when the diode has a nominal current of 75 ma flowing through it.

3.5.2 Transistors (Q1 through Q8).

3.5.2.1 Direct current gain.-

(a) The direct-current gain (B) shall be 28.4 minimum, when the base current (I_B) is 1.7 ma (max) and the collector current (I_C) is 48.3 ma (min).

(b) The direct-current gain (B) shall be 17.2 minimum, when the base current (I_B) is 55.0 ma (max) and the collector current (I_C) is 9.45 ma (min).

3.5.2.2 Reverse current.-

(a) The leakage current (I_{CO}) shall be less than 1.5 ma, when collector-to-base voltage (V_{CB}) is 60 Vdc ± 2%.

(b) The leakage current (I_{CO}) shall be less than 250 ua, when collector-to-base voltage (V_{CB}) is 30 Vdc ± 2%.

3.5.2.3 DC saturation voltage (V_{CE}) sat).- The DC saturation voltage (V_{CE}) sat) shall be less than .30 Vdc, when I_C) is 900 ma ± 2% and I_B is 100 ma ± 2%.

3.5.3 Extreme operating temperatures (-40°F and +135°F).

3.5.3.1 Diodes.

3.5.3.1.1 Reverse current.- The leakage current (I_R) through diodes Cr 1, 2, 10, 11, 18, 20, 21 and 23, shall be less than 12.0 ma, when each diode is back-biased with 150 Vdc ± 2%.

3.5.3.1.2 Reverse current.- The leakage current (I_R) through diodes CR 3, 4, 5, 6, 7, 8, 12, 13, 14, 15, 16, 17, 24, 25, 26, 27, 28 and 29, shall be less than 1.8 ma, when each diode is back-biased with 250 Vdc ± 2%.

3.5.3.1.3 Forward voltage drop.- The forward voltage drop (V_F) across diodes 1, 2, 10, 11, 18, 20, 21 and 23, shall be less than 1.5 Vdc, when each diode has a forward current (I_F) of 1 amp, nominal.

3.5.3.1.4 Forward voltage drop.- The forward voltage drop (V_F) across diodes CR 3, 4, 5, 6, 7, 8, 12, 13, 14, 15, 16, 17, 24, 25, 26, 27, 28 and 29, shall be less than 1.5 Vdc, when each diode has a forward current (I_F) of 200 ma (nominal).

3.5.3.1.5 Zener voltages (CR 19 and CR 22).- The zener voltage (V_Z) of diodes CR19 and CR22 shall be 6.0 ± 0.36 Vdc, when each diode has a nominal current of 250 ma flowing through it.

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3.5.3.1.6 Zener voltage (CR 9).- The zener voltage (V_Z) of diode CR9 shall be 33 ± 6 Vdc, when the diode has a nominal current of 75 ma flowing through it.

3.5.3.2 Transistors (Q1 through Q8).

3.5.3.2.1 Direct current gain.-

(a) The direct-current gain (B) shall be 12.2 (min), when the base current (I_B) is 3.8 ma (max) and the collector current (I_C) is 46.2 ma (min).

(b) The direct-current gain (B) shall be 8.1 (min), when the base current (I_B) is 110 ma (max) and the collector current (I_C) is 890 ma (min).

3.5.3.2.2 Reverse current.- The leakage current (I_{CO}) shall be less than 3.0 ma, when collector-to-base voltage (V_{CB}) is 60 Vdc ± 2%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of Product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspections provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

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TABLE I
(100 percent inspection)

Characteristic	Requirements	Test Procedures
101. Isolation resistance to heat sink	3.4	4.5.2.1
102. Reverse Current (diodes CR1, 2, 10, 11, 18, 20, 21, 23)	3.5.1.1	4.5.2.2.1
103. Reverse current (diodes CR 3 to 8, 12 to 17, 24 to 29)	3.5.1.2	4.5.2.2.2
104. Forward Voltage drop (diodes CR1, 2, 10, 11, 18, 20, 21, 23)	3.5.1.3	4.5.2.2.3
105. Forward Voltage drop (diodes CR 3 to 8, 12 to 17, 24 to 29)	3.5.1.4	4.5.2.2.4
106. Zener Voltage (CR 19 and 22)	3.5.1.5	4.5.2.2.5
107. Zener Voltage (CR 9)	3.5.1.6	4.5.2.2.6
108. D.C. Gain (Q1 to Q8) (28.4 min)	3.5.2.1 (a)	4.5.2.3.1
109. D.C. Gain (Q1 to Q8) (17.2 min)	3.5.2.1 (b)	4.5.2.3.2
110. Reverse current (Q1 to Q8) (60Vdc)	3.5.2.2 (a)	4.5.2.3.3
111. Reverse current (Q1 to Q8) (30 Vdc)	3.5.2.2 (b)	4.5.2.3.4
112. Saturation Voltage	3.5.2.3	4.5.2.3.5

4.3 Control tests.

4.3.1 General.- One assembly shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings - visual
302. General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table III.

TABLE III

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303. Vibration	3.3.1	4.5.1.1
304. Shock	3.3.2	4.5.1.2
305. Humidity-Operating	3.3.4.1	4.5.1.3.2
306. Humidity-Non-operating	3.3.4.2	4.5.1.3.1
307. Salt spray	3.3.6	4.5.1.4
308. Operating temperature	3.3.3.1, 3.5.3.1, 3.5.3.2	4.5.1.5, 4.5.2.4.1, 4.5.2.4.2
309. Storage temperature	3.3.3.2	4.5.1.5

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4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (See 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the current and voltage supplies in Tables IV and V shall be utilized in the performance of the applicable tests as specified in 4.5:

TABLE IV

D.C. VOLTAGE SUPPLIES

Voltage Supply Number	Nominal Voltage (volts)	Voltage Tolerance (percent)	Max. RMS Ripple	Current Requirement	D.C. Output Impedance Maximum (ohms)
1	150	± 2	0.5 volts	0-12 ma	500
2	250	± 2	1.0 volts	0-2 ma	6K
3	60	± 2	0.100 volts	0-3 ma	1K
4	30	± 2	0.100 volts	0-250 ua	6K
5	.5	± 2	0.010 volts	0-1.5 amp	15×10^{-3}

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TABLE V

D.C. CURRENT SUPPLIES

Current Supply Number	Nominal Current	Current Tolerance (percent)	Max. RMS Ripple (ma)	Voltage Requirement (volts)	D.C. Output Admittance Maximum (ohms)
1	1 amp	± 2	10	0-2.0	2.5×10^{-3}
2	200 ma	± 2	5	0-2.0	5×10^{-3}
3	.5 amp	± 2	10	4.5-7	8×10^{-3}
4	75 ma	± 2	2	26-37	$.2 \times 10^{-3}$
5	50 ma	± 2	.2	0-2	1.5×10^{-3}
6	100 ma	± 2	1	0-2	2.5×10^{-3}

4.4.1.1.1 Additional test equipment shall include the following:

Item	Accuracy
1. Voltmeter	D.C. 0 to 50 volts input impedance greater than 1 megohm.
2. Microamp-meter (D.C.)	Accuracy of scale reading between 30 and 80% of the scale range to be within 2%.
3. Milliammeter (D.C.)	Accuracy of scale readings between 30 and 80% of the scale range to be within 2%.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the assembly rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.5.

4.5,1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the rectifier shall be shocked in the vertically plane only. The impact (shocks and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector re-
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ceptacle that receives the rectifier in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.5 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 $\pm$ 5 percent for the 24 hour period and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.9 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The assembly shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2.4 and 4.5.2.5 for compliance with 3.3.6.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the assembly shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.2.9.1 and 4.5.2.9.2 to determine compliance with the requirements of 3.5.3 as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2.4 and 4.5.2.5 inclusive to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 assemblies produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance.

4.5.2.1 Isolation resistance to heat sinks.- Compliance with 3.4 shall be determined by using voltage supply No. 1 specified in Table IV. Connections and tests on the rectifier transistor assembly shall be performed as indicated in Table VI and current drawn shall be no more than shown in the maximum current column. Remove all power supplies and test leads from the assembly.

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TABLE VI

ISOLATION TEST

Connect Voltage Supply (No. 1)		
Positive Terminal to Rectifier Transistor Assembly Terminal	Negative Terminal to Rectifier Transistor Assembly Terminal	Maximum Current Milliamperes
41A	46A	6
41A, 46A	52A	6
36A	51A	.025
36A, 51A	35A	.025
36A, 51A, 35A	36B	.025
36A, 51A, 35A, 36B	46B	.025
6A	25A	.025
6A, 25A	11A	.025
6A, 25A, 11A	26A	.025
4B	10B	.025
4B, 10B	28B	.025
4B, 10B, 28B	19A	.025

4.5.2.2 Diodes.

4.5.2.2.1 Reverse current (CR 1, 2, 10, 11, 18, 20, 21, 23).- Compliance with 3.5.1.1 shall be determined by connecting the voltage supply No. 1 as specified in Table IV to test terminals X and Y as specified in Table VII. The negative polarity shall be connected to test terminal X and the positive polarity shall be connected to the test terminal Y in each numbered test position. For each test position, the current drawn shall be less than 6 milliamperes (ma) as specified in 3.5.1.1. Remove all power supplies and test leads from the assembly.

4.5.2.2.2 Reverse current (CR 3 to 8, 12 to 17, 24 to 29).- Compliance with 3.5.1.2 shall be determined by connecting the voltage supply No. 2 of Table IV to test terminals X and Y as specified in Table VIII. The negative polarity shall be connected to test terminal X and the positive polarity shall be connected to test terminal Y in each numbered test position. For each test position, the current drawn shall be less than 36 microamperes (ua) as specified in 3.5.1.2. Remove all power supplies and test leads from the assembly.

4.5.2.2.3 Forward voltage drop (CR 1, 2, 10, 11, 18, 20, 21, 23).- Compliance with 3.5.1.3 shall be determined by connecting the current supply No. 1 of Table V to test terminals X and Y as specified in Table VII. The current shall flow into test terminal X and out of test terminal Y in each numbered test position. For each test position, the voltage drop across test terminals X and Y shall be less than 1.5 volts as specified in 3.5.1.3. Remove all power supplies and test leads from the assembly. The direction of the

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electric current is taken as the direction in which positive electricity flows.

4.5.2.2.4 Forward voltage drop (CR 3 to 8, 12 to 17, 24 to 29).- Compliance with 3.5.1.4 shall be determined by connecting the current supply No. 2 of Table V to test terminals X and Y as specified in Table VIII. The current shall flow into test terminal X and out of test terminal Y in each numbered test position. For each position, the voltage drop across test terminals X and Y shall be less than +1.5 volts as specified in 3.5.1.4. Remove all power supplies and test leads from the assembly.

4.5.2.2.5 Zener (CR 19 and 22).- Compliance with 3.5.1.5 shall be determined by connecting the current supply No. 3 of Table V to terminals 36B and 41B of the Rectifier Transistor Assembly such that current flows into terminal 36B and out of 41B. The voltage across terminal 36B and 41B shall be +6.0 ± 0.36 volts. Repeat the above test applying current into 46B and out of 52B. The voltage across terminals 46B and 52B shall be +6.0 ± 0.36 volts as specified in 3.5.1.5. Remove all power supplies and test leads from the assembly.

4.5.2.2.6 Zener (CR 9) voltage.- Compliance with 3.5.1.6 shall be determined by connecting the current supply No. 4 of Table V to terminals 35B and 35A of the Rectifier Transistor Assembly such that the current flows into terminal 35B and out of 35A. The voltage across 35B and 35A shall be 33 ± 2 volts as specified in 3.5.1.6. Remove all power supplies and test leads from the assembly.

4.5.2.3 Transistors (Q1 to Q8).

4.5.2.3.1 D.C. gain (28.4 min).- Compliance with 3.5.2.1(a) shall be determined by connecting the following equipment in the order given, terminals X, Y and Z given in Table IX.

(a) Current measuring device of less than 50 ohms resistance, from test terminal X to common.

(b) Voltage supply No. 5 as specified in Table IV to test terminal Y such that the negative polarity is connected to Y and the positive polarity is connected to common. Energize voltage supply.

(c) Current supply No. 5, Table V, to test terminal Z such that current flows into Z and out of common.

4.5.2.3.1.1 The current out of terminal X as measured at each of the eight test positions shall be a maximum of 1.7 milliamperes to determine conformance with 3.5.2.1(a). Remove all power supplies and test leads from the assembly.

4.5.2.3.2 D.C. gain (17.2 min).- Compliance with 3.5.2.1(b) shall be determined by connecting the following equipment in the order given to terminals X, Y and Z given in Table IX:

(a) Current measuring device of less than 2 ohms resistance from test terminal X to common.

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(b) Voltage supply No. 5 as specified in Table IV to test terminal Y such that the negative polarity is connected to Y and the positive polarity is connected to common.

(c) Current supply No. 1, Table V, to test terminal Z and common such that current flows into Z and out of the common.

4.5.2.3.2.1 The current out of terminal X as measured at each of the eight test positions shall be a maximum of +55 milliamperes to determine compliance with 3.5.2.1(b). Remove all power supplies and test leads from the assembly.

TABLE VII

MEDIUM CURRENT SILICON RECTIFIER TEST TERMINALS

Test Position	Test Terminal X Rectifier Transistor Assembly Terminals	Test Terminal Y Rectifier Transistor Assembly Terminals	Diode Tested
1	52A, 54A	36A, 37A	CR 20
2	46A, 48A	36A, 37A	CR 10
3	41A, 43A	36A, 37A	CR 1
4	63A, 64A	36A, 37A	CR 23
5	57A, 57B	52A, 54A	CR 21
6	57A, 57B	46A, 48A	CR 11
7	57A, 57B	41A, 43A	CR 2
8	61B, 62A	51A, 51B	CR 18

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TABLE VIII

LOW CURRENT SILICON RECTIFIER TEST TERMINALS

Test Position	Test Terminal X Rectifier Transistor Assembly Terminals	Test Terminal Y Rectifier Transistor Assembly Terminals	Diode being Tested
1	11A, 12A, 13A	1B, 2B, 3B	CR 24
2	11B, 12B, 13B	1B, 2B, 3B	CR 12
3	1A, 2A	1B, 2B, 3B	CR 3
4	14A, 14B, 15A, 15B	11A, 12A, 13A	CR 25
5	14A, 14B, 15A, 15B	11B, 12B, 13B	CR 13
6	14A, 14B, 15A, 15B	1A, 2A, 3A	CR 4
7	24A, 24B	17B, 18B, 19B	CR 26
8	20B, 22B	17B, 18B, 19B	CR 14
9	17A, 18A	17B, 18B, 19B	CR 5
10	29B, 30A, 30B	24A, 24B	CR 27
11	29B, 30A, 30B	20B, 22B	CR 15
12	29B, 30A, 30B	17A, 18A	CR 6
13	62B, 63B	31B, 32B	CR 28
14	33B, 34B	31B, 32B	CR 16
15	33A, 34A	31B, 32B	CR 7
16	31A, 32A	62B, 63B	CR 29
17	31A, 32A	33B, 34B	CR 17
18	31A, 32A	33A, 34A	CR 8

4.5.2.3.3 Reverse current (60 Vdc).- Compliance with 3.5.2.2(a) shall be determined by connecting voltage supply No. 3 Table IV to test terminals X and Y as specified in Table IX. The positive polarity shall be connected to test terminal X and the negative polarity shall be connected to test terminal Y in each numbered test position. For each test position, the current drawn shall be less than 1.5 ma as specified in 3.5.2.2(a). Remove all power supplies and test leads from the assembly.

4.5.2.3.4 Reverse current (30 Vdc).- Compliance with 3.5.2.2(b) shall be determined by connecting voltage supply No. 4 Table IV to test terminals X and Y as specified in Table IX. The positive polarity shall be connected to test terminal X and the negative polarity shall be connected to test terminal Y in each numbered test position. For each test position, the current drawn shall be less than 250 microamperes as specified in 3.5.2.2(b). Remove all power supplies and test leads from the assembly.

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4.5.2.3.5 D.C. Saturation voltage.- Compliance with 3.5.2.3 shall be determined by connecting the following equipment in the order given to test terminals X, Y and Z given in Table VI. Measure the voltage at each of the eight test positions.

(a) Terminal Y to common.

(b) Current supply No. 6 of Table V to terminal X and common such that current flows out of X and into the common.

(c) Current supply No. 1 as specified in Table V to test terminal Z and common such that current flows into Z and out of the common.

4.5.2.3.6 The measured voltage across test terminals Z and Y, with Z as the positive terminal, shall be less than +0.30 volts dc as specified in 3.5.2.3. Remove all power supplies and test leads from the assembly.

4.5.2.4 Extreme operating temperatures (-40°F and +135°F).- The following tests shall be conducted in conformance with 3.3.3.1 to determine compliance with 3.5.3.

4.5.2.4.1 Diodes.

4.5.2.4.1.1 Reverse current (CR 1, 2, 10, 11, 18, 20, 21, 23).- Repeat tests specified in 4.5.2.2.1 except measure the current drawn to be less than 12.0 milliamperes as specified in 3.5.3.1.1.

4.5.2.4.1.2 Reverse current (CR 3 to 8, 12 to 17, 24 to 29).- Repeat tests specified in 4.5.2.2.2 except measure the current drawn to be less than 1.8 milliamperes as specified in 3.5.3.1.2.

4.5.2.4.1.3 Forward voltage drop (CR 1, 2, 10, 11, 18, 20, 21, 23).- Repeat the test specified in 4.5.2.2.3. Test results shall be as specified in 3.5.1.3.3.

4.5.2.4.1.4 Forward voltage drop (CR 3 to 8, 12 to 17, 24 to 29).- Repeat the test specified in 4.5.2.2.4. Test results shall be as specified in 3.5.1.3.4.

4.5.2.4.1.5 Zener voltage (CR 19-22).- Repeat the test specified in 4.4.2.2.5. Test results shall conform to the requirements of 3.5.3.1.5.

4.5.2.4.1.6 Zener voltage (CR 9).- Repeat the test specified in 4.5.2.2.6 except across terminals 35A and 35B of the Rectifier Transistor Assembly measure +33 ± 6 volts as required in 3.5.3.1.6.

4.5.2.4.2 Transistors (Q1 to Q8).

4.5.2.4.2.1 Reverse current (60 Vdc).- Repeat tests specified in 4.5.2.3.3 except the current drawn shall be less than 3 milliamperes as specified in 3.5.3.2.2.

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4.5.2.4.2.2 D.C. gain (12.2 min).- Repeat tests specified in 4.5.2.3.1 except the current shall be less than 3.8 milliamperes out of test terminal X as specified in 3.5.3.2.1(a).

4.5.2.4.2.3 D.C. gain (8.1 min).- Repeat test specified in 4.5.2.3.2 except the current shall be less than 110 milliamperes out of test terminal X as specified in 3.5.3.2.1(b).

TABLE IX

TRANSISTOR TEST TERMINALS

Test Position	Test Terminal X Rectifier Transistor Assembly Terminals	Test Terminal Y Rectifier Transistor Assembly Terminals	Test Terminal Z Rectifier Transistor Assembly Terminals	Transistor being tested
1	6B	6A	5A	Q5
2	5B	4B	4A	Q4
3	8B	10B	7B	Q1
4	23A	25A	22A	Q6
5	8A	10A	7A	Q2
6	26B	28B	25B	Q7
7	23B	19A	20A	Q3
8	29A	26A	28A	Q8

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

6. NOTES

6.1 Intended use.- The rectifier transistor assembly houses the power rectifier used to convert three phase voltage into D.C. voltage and the medium power transistors used as drivers for several of the regulated power supplies. Zener diodes, mounted on the Diode, Rectifier Assembly, regulate bias levels used in the computer.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet (See 5.1).
(e) A block of serial numbers to identify each rectifier.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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